

The CONTENTS Procedure

Data Set Name	WORK.IMPORT	Observations	440833
Member Type	DATA	Variables	12
Engine	V9	Indexes	0
Created	05/09/2026 00:52:34	Observation Length	96
Last Modified	05/09/2026 00:52:34	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	324
First Data Page	1
Max Obs per Page	1363
Obs in First Data Page	1327
Number of Data Set Repairs	0
Filename	/saswork/SAS_workB5F70000EF99_odaws02-usw2-2.oda.sas.com/SAS_work6F130000EF99_odaws02-usw2-2.oda.sas.com/import.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2281705494
Access Permission	rw-r--r--
Owner Name	u64508356
File Size	41MB
File Size (bytes)	42598400

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
2	Age	Num	8	BEST12.	BEST32.
12	Churn	Num	8	BEST12.	BEST32.
9	Contract Length	Char	9	\$9.	\$9.
1	CustomerID	Num	8	BEST12.	BEST32.
3	Gender	Char	6	\$6.	\$6.
11	Last Interaction	Num	8	BEST12.	BEST32.
7	Payment Delay	Num	8	BEST12.	BEST32.
8	Subscription Type	Char	8	\$8.	\$8.
6	Support Calls	Num	8	BEST12.	BEST32.
4	Tenure	Num	8	BEST12.	BEST32.
10	Total Spend	Num	8	BEST12.	BEST32.
5	Usage Frequency	Num	8	BEST12.	BEST32.

First 10 Rows of the Dataset (Head)

Obs	CustomerID	Age	Gender	Tenure	Usage Frequency	Support Calls	Payment Delay	Subscription Type	Contract Length	Total Spend	Last Interaction	Churn
1	2	30	Female	39	14	5	18	Standard	Annual	932	17	1
2	3	65	Female	49	1	10	8	Basic	Monthly	557	6	1
3	4	55	Female	14	4	6	18	Basic	Quarterly	185	3	1
4	5	58	Male	38	21	7	7	Standard	Monthly	396	29	1
5	6	23	Male	32	20	5	8	Basic	Monthly	617	20	1
6	8	51	Male	33	25	9	26	Premium	Annual	129	8	1
7	9	58	Female	49	12	3	16	Standard	Quarterly	821	24	1
8	10	55	Female	37	8	4	15	Premium	Annual	445	30	1
9	11	39	Male	12	5	7	4	Standard	Quarterly	969	13	1
10	12	64	Female	3	25	2	11	Standard	Quarterly	415	29	1

Frequency Distribution of Categorical Variables

The FREQ Procedure

Gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	190580	43.23	190580	43.23
Male	250252	56.77	440832	100.00
Frequency Missing = 1				

Subscription Type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Basic	143026	32.44	143026	32.44
Premium	148678	33.73	291704	66.17
Standard	149128	33.83	440832	100.00

Subscription Type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Frequency Missing = 1				

Contract Length	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Annual	177198	40.20	177198	40.20
Monthly	87104	19.76	264302	59.96
Quarterly	176530	40.04	440832	100.00
Frequency Missing = 1				

Churn	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	190833	43.29	190833	43.29
1	249999	56.71	440832	100.00
Frequency Missing = 1				

Missing values in numeric columns

The MEANS Procedure

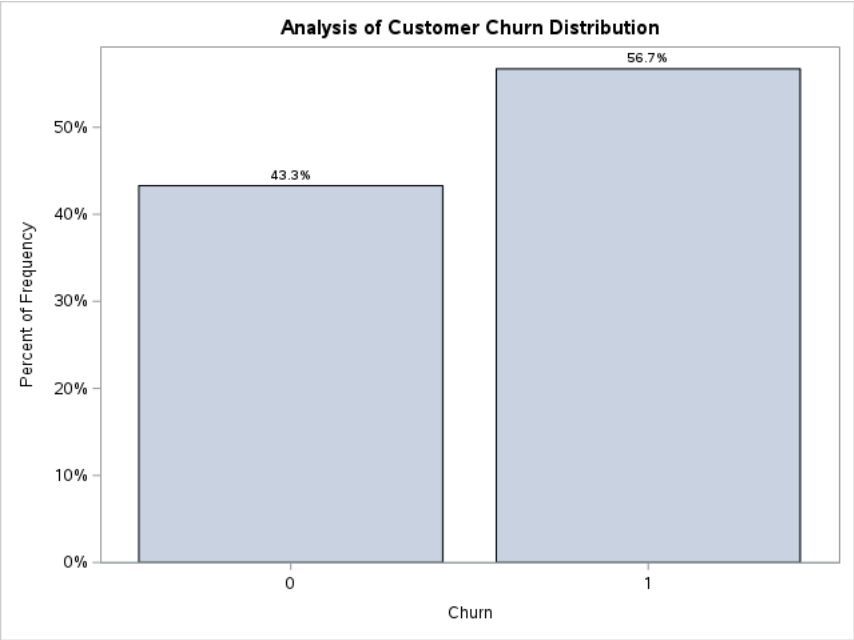
Variable	N	N Miss
CustomerID	440832	1
Age	440832	1
Tenure	440832	1
Usage Frequency	440832	1
Support Calls	440832	1
Payment Delay	440832	1
Total Spend	440832	1
Last Interaction	440832	1
Churn	440832	1

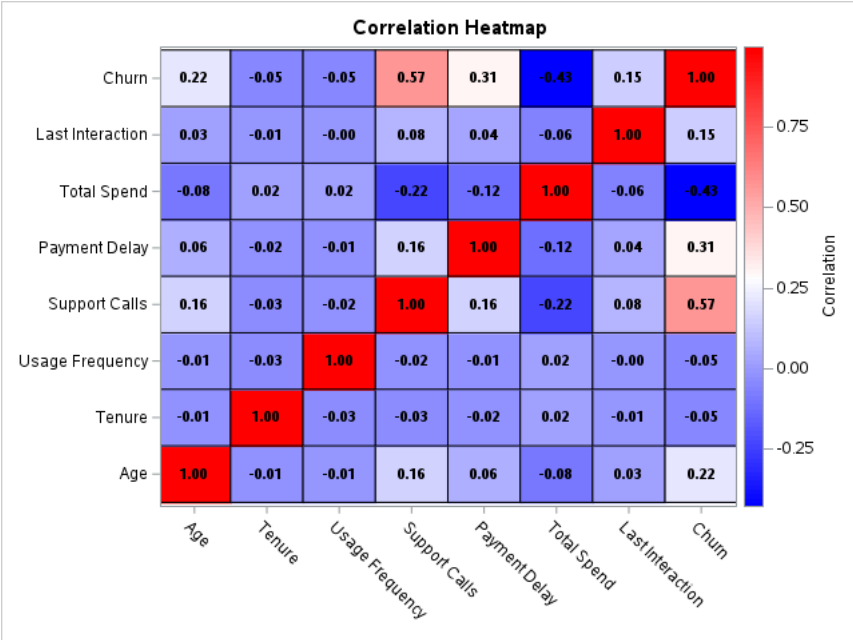
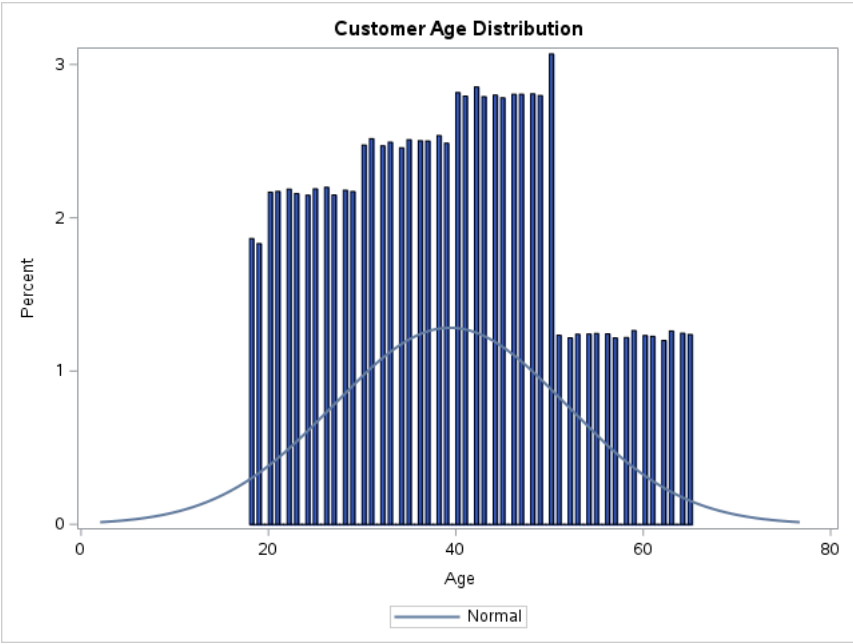
The MEANS Procedure

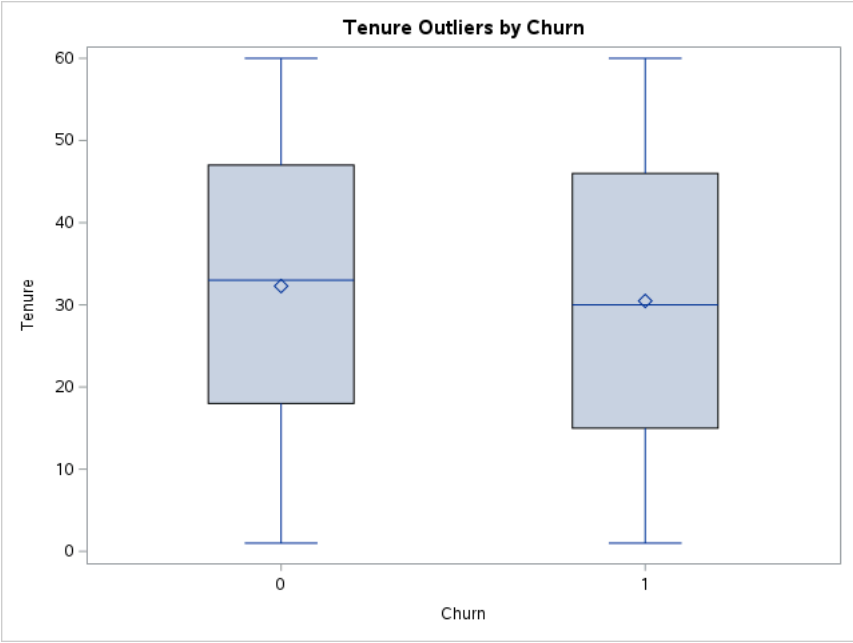
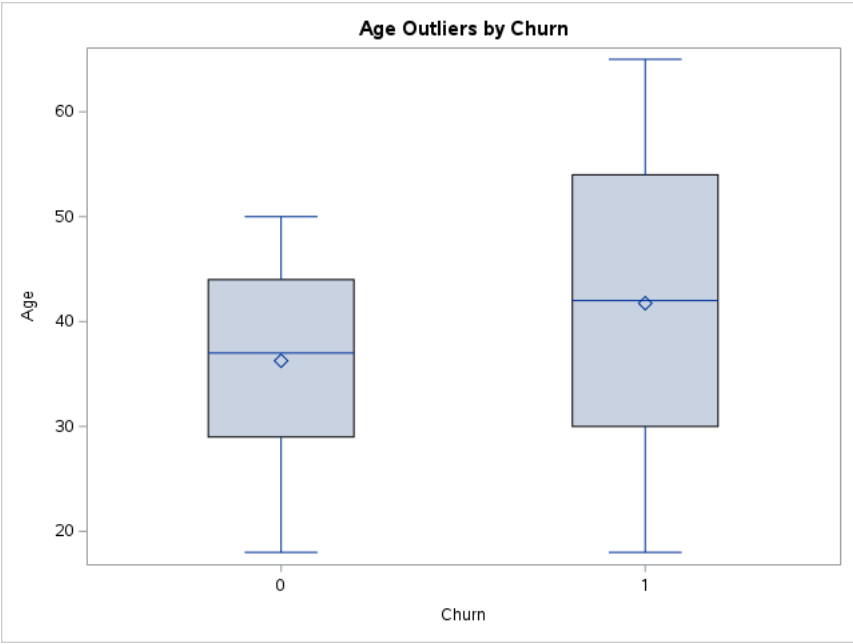
Variable	N	Mean	Std Dev	Minimum	Maximum
Age	440832	39.3731535	12.4423695	18.0000000	65.0000000
Tenure	440832	31.2563357	17.2557274	1.0000000	60.0000000
Usage Frequency	440832	15.8074936	8.5862416	1.0000000	30.0000000
Support Calls	440832	3.6044366	3.0702179	0	10.0000000
Payment Delay	440832	12.9657216	8.2580625	0	30.0000000
Total Spend	440832	631.6162228	240.8030011	100.0000000	1000.00
Last Interaction	440832	14.4808680	8.5962077	1.0000000	30.0000000

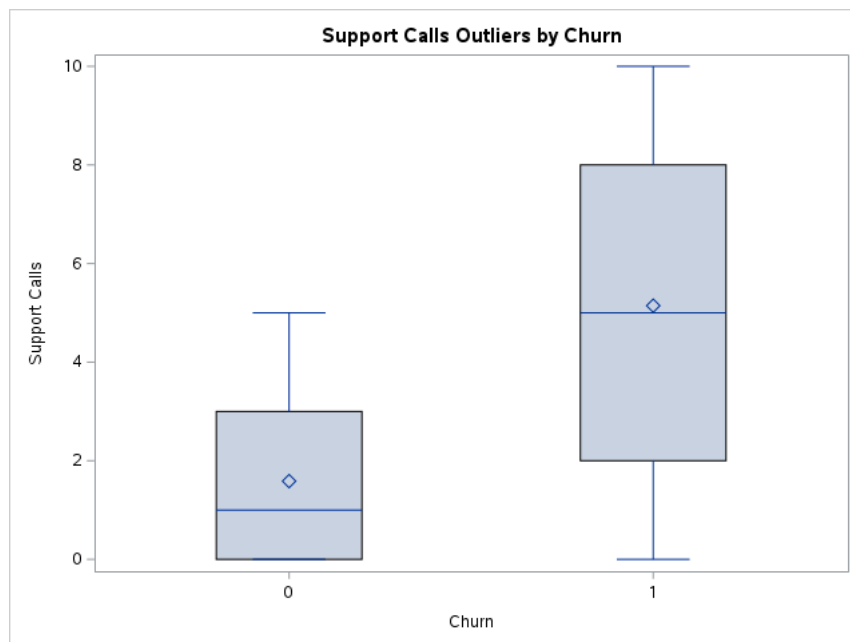
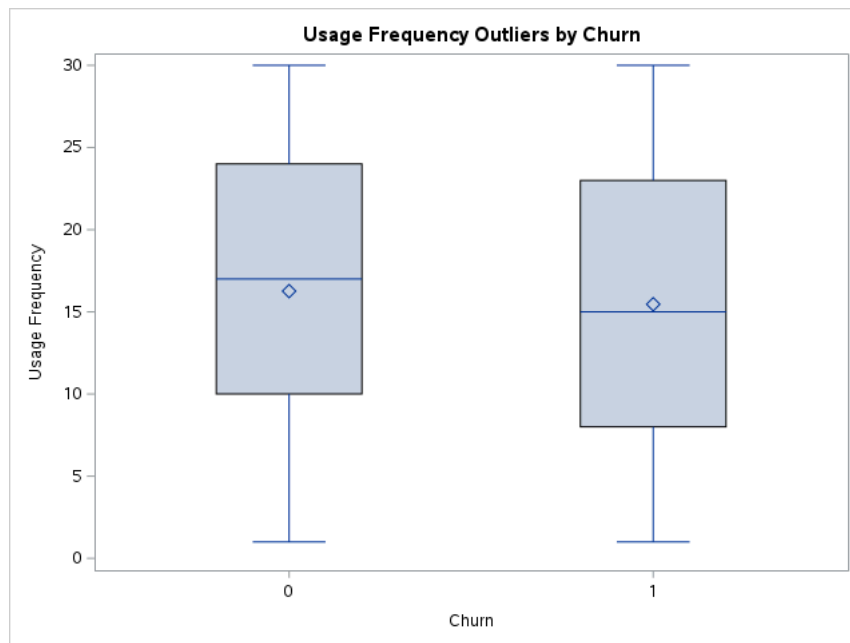
Total Number of Duplicate Customer IDs

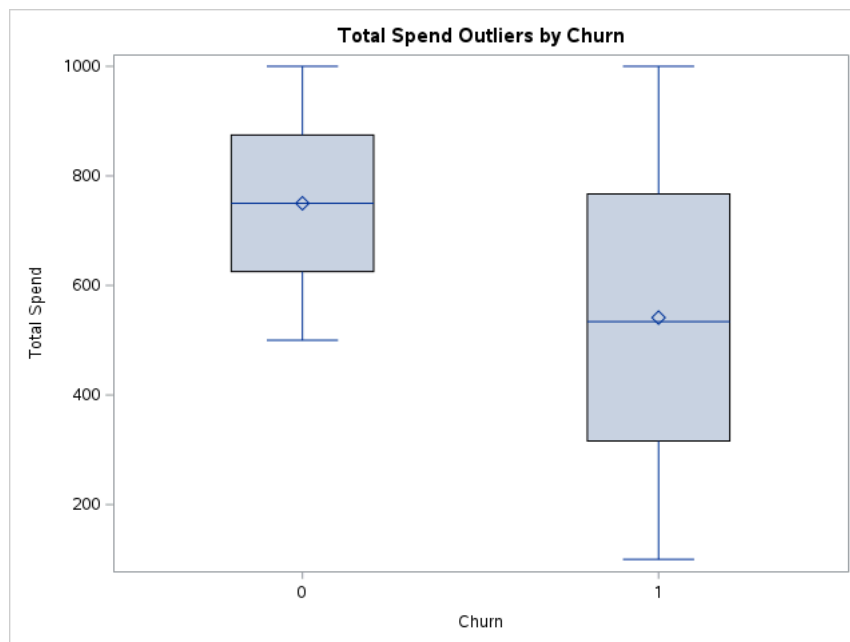
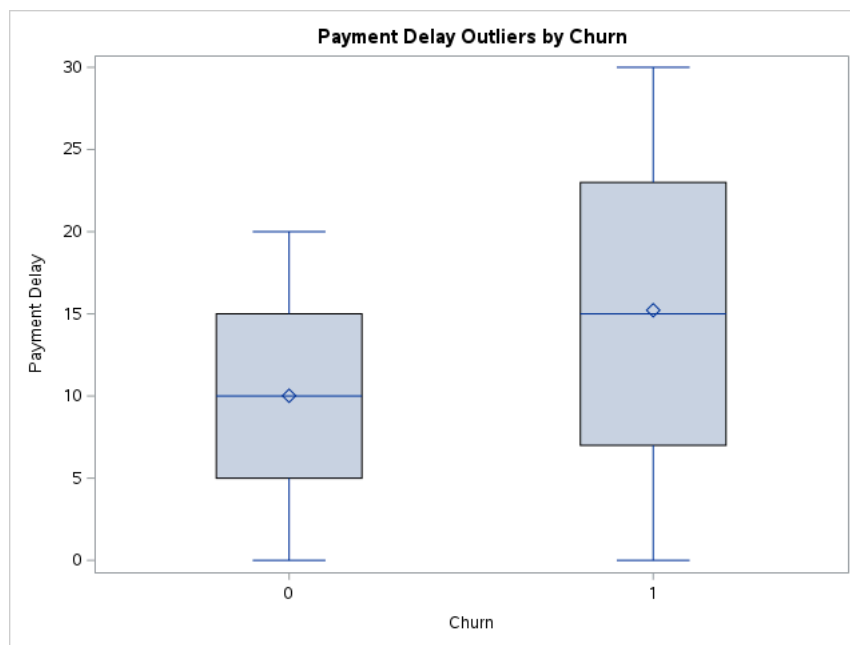
Duplicates
0

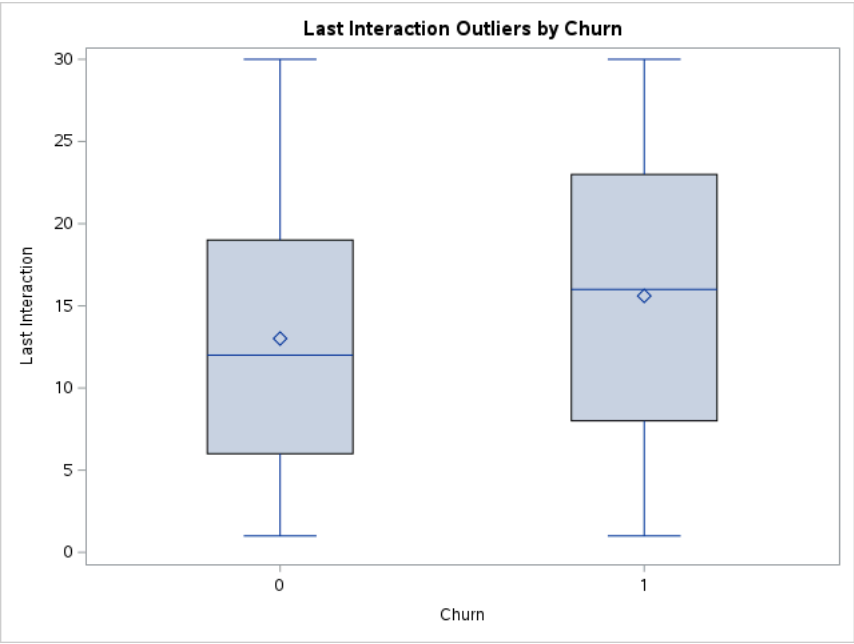












Last Interaction Outliers by Churn

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
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Input Data Set	IMPORT
Random Number Seed	42
Sampling Rate	0.8
Sample Size	352667
Selection Probability	0.800001
Sampling Weight	0
Output Data Set	TRAIN_SELECTED

Train Set Size

Train_Rows
352667

Test Set Size

Test_Rows
88166

The MEANS Procedure

Variable	N	N Miss
CustomerID	352667	0
Age	352667	0
Tenure	352667	0
Usage Frequency	352667	0
Support Calls	352667	0
Payment Delay	352667	0
Total Spend	352667	0
Last Interaction	352667	0
Churn	352667	0

The MEANS Procedure

Variable	N	N Miss
CustomerID	88165	1
Age	88166	0
Tenure	88165	1
Usage Frequency	88165	1
Support Calls	88165	1
Payment Delay	88165	1
Total Spend	88166	0
Last Interaction	88165	1
Churn	88165	1

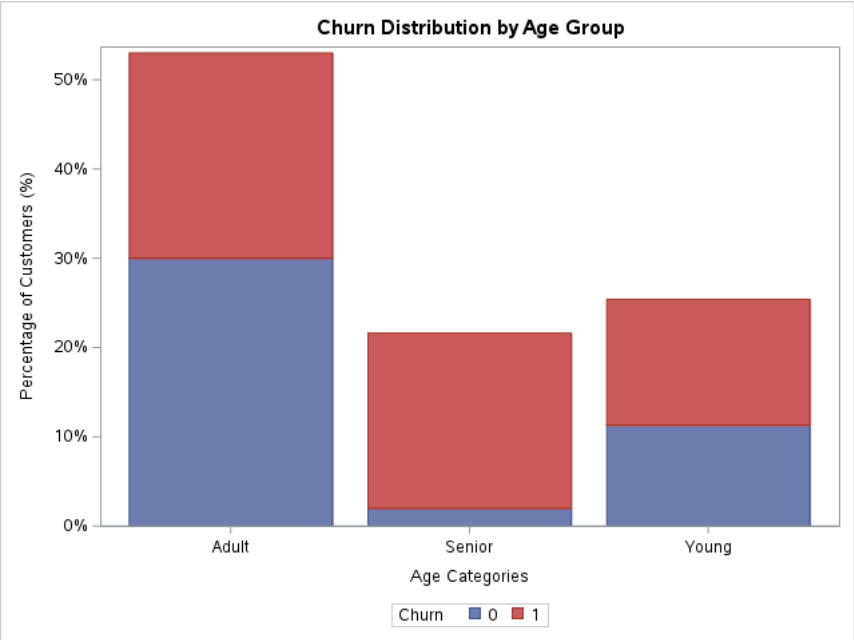
Preview of the 4 New Engineered Features

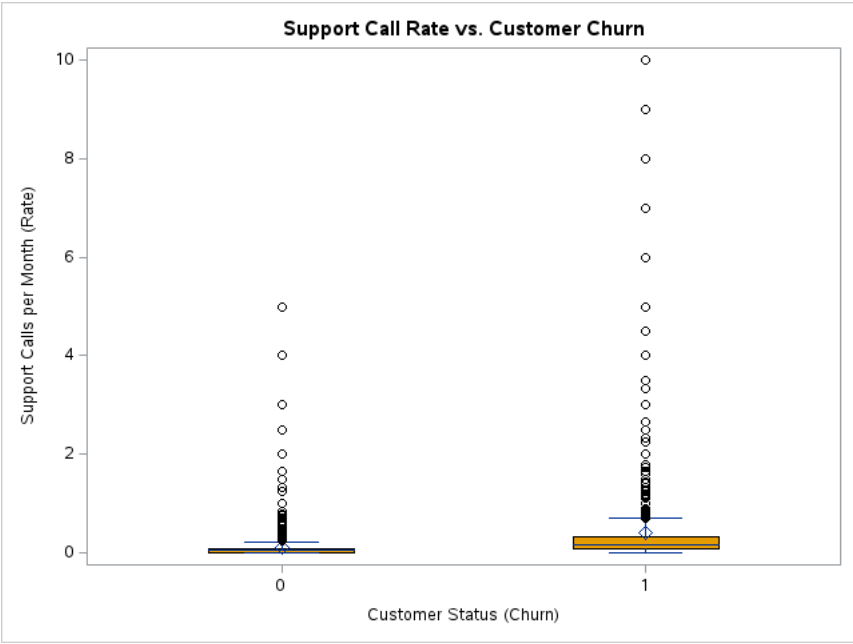
Obs	CustomerID	SupportCallRate	SpendingPerTenure	PaymentRisk	AgeGroup
1	1	0.12821	23.897	23	Adult
2	2	0.20408	11.367	18	Senior
3	3	0.18421	10.421	14	Senior
4	4	0.15625	19.281	13	Young
5	5	0.27273	3.909	35	Senior
6	6	0.10811	12.027	19	Senior
7	7	0.58333	80.750	11	Adult
8	8	0.66667	138.333	13	Senior
9	9	0.00000	51.667	30	Young
10	10	0.24390	6.463	35	Young
11	11	0.02857	14.800	14	Adult
12	12	0.18421	25.737	18	Young
13	13	0.02439	3.683	6	Adult
14	14	0.11364	15.205	9	Young
15	15	0.13333	17.467	16	Adult

Statistical Summary for New Features

The MEANS Procedure

Variable	N	Mean	Minimum	Maximum	Std Dev
SupportCallRate	352667	0.2741983	0	10.0000000	0.7232847
SpendingPerTenure	352667	45.9742943	1.6666667	1000.00	93.1924738
PaymentRisk	352667	16.5678303	0	40.0000000	9.2606331





First 10 Rows of TRAIN Dataset

Obs	CustomerID	Age	Tenure	Usage Frequency	Support Calls	Payment Delay	Total Spend	Last Interaction	Churn	Gender_Female	Gender_Male	Gender_Unknown	SubType_Enc	Contract_Enc
1	1	-0.753610041	0.4479529652	-0.210385605	0.4552629388	0.6097723842	1.2484042268	0.2922485931	1	1	0	0	1	2
2	2	2.0598777646	1.0272597268	-1.723939876	2.0831004153	-0.601583034	-0.309481582	-0.988143409	1	1	0	0	0	0
3	3	1.4971802034	0.390022289	0.6046051554	1.1063979294	-0.722718576	-0.97833389	1.689039868	1	0	1	0	1	0
4	4	-1.316307603	0.042438232	0.4881779039	0.4552629388	-0.601583034	-0.060219853	0.6414464118	1	0	1	0	0	0
5	5	0.9344826422	0.1003689082	1.0703141616	1.75753292	1.5788567187	-2.087548586	-0.755344863	1	0	1	0	2	2
6	6	1.2560241057	0.3320916129	-0.908949115	0.1296954435	0.2463657588	-0.774770144	1.8054391409	1	1	0	0	2	2
7	7	-0.030141748	-1.116175291	-1.258230869	1.1063979294	-1.086125201	1.4021156266	-0.173348499	1	0	1	0	1	1
8	8	1.9794923987	-1.637551377	1.0703141616	-0.521439547	-0.238176408	-0.899401009	1.689039868	1	1	0	0	1	1
9	9	-0.833995407	-0.768591234	-0.792521863	-1.172574538	2.0633988859	1.2400955025	0.408647866	1	0	1	0	2	1
10	10	-1.396692969	0.5638143175	0.1388961492	2.0831004153	1.4577211769	-1.522555333	0.9906442306	1	0	1	0	0	1

First 10 Rows of TEST Dataset

Obs	CustomerID	Age	Tenure	Usage Frequency	Support Calls	Payment Delay	Total Spend	Last Interaction	Churn	Gender_Female	Gender_Male	Gender_Unknown	SubType_Enc	Contract_Enc
1	1	1.2560241057	-1.000313939	-1.374658121	0.7808304341	0.6097723842	-1.854904305	-1.337341228	1	1	0	0	0	1
2	2	1.4971802034	1.0272597268	-0.443240109	-0.195872052	0.3675013006	0.7872700273	1.1070435035	1	1	0	0	1	1
3	3	1.0148680081	-0.594799206	-1.141803618	-0.195872052	1.5788567187	0.8246592867	0.5250471389	1	1	0	0	2	0
4	4	-1.235922237	-1.579620701	-0.792521863	0.1296954435	1.0943145514	-1.775971424	-1.220941955	1	0	1	0	1	1
5	5	0.7737119104	1.43277446	0.1388961492	-0.521439547	2.0633988859	1.4270417996	0.2922485931	1	0	1	0	1	1
6	6	1.4167948375	1.3748437838	1.3031686647	-0.195872052	-1.207260743	-1.742736527	-0.522546317	1	1	0	0	0	2
7	7	-0.994766139	1.2010517553	-0.908949115	1.1063979294	-1.207260743	-0.820468128	0.5250471389	1	1	0	0	1	0
8	8	-1.718234432	0.3320916129	-0.093958354	1.4319654247	-0.843854118	0.700028422	1.689039868	1	0	1	0	2	0
9	9	-0.833995407	0.6796756699	-0.443240109	-0.521439547	-0.35931195	-1.422850641	-0.988143409	1	0	1	0	2	2
10	10	1.8991070328	1.1431210792	-1.491085372	0.4552629388	-0.117040867	-1.427005003	0.0594500473	1	1	0	0	1	2

The LOGISTIC Procedure

Model Information	
Data Set	WORK.TRAIN
Response Variable	Churn
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	352667
Number of Observations Used	352667

Response Profile		
Ordered Value	Churn	Total Frequency
1	1	199961
2	0	152706

Probability modeled is Churn='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	482551.34	213791.92
SC	482562.11	213931.98
-2 Log L	482549.34	213765.92

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	268783.415	12	<.0001
Score	185517.281	12	<.0001
Wald	84504.8439	12	<.0001

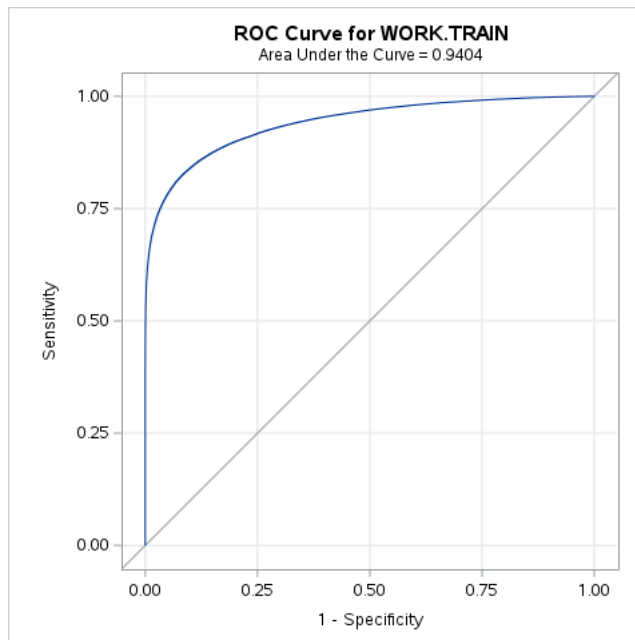
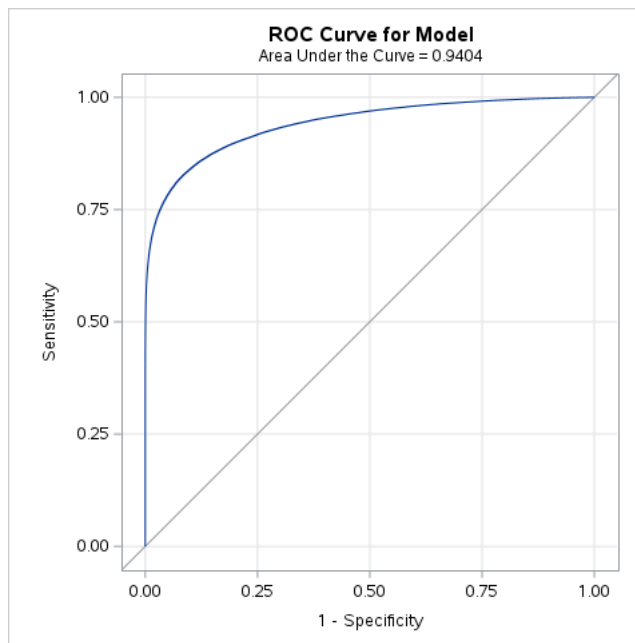
Note: The following parameters have been set to 0, since the variables are a linear combination of other variables as shown.

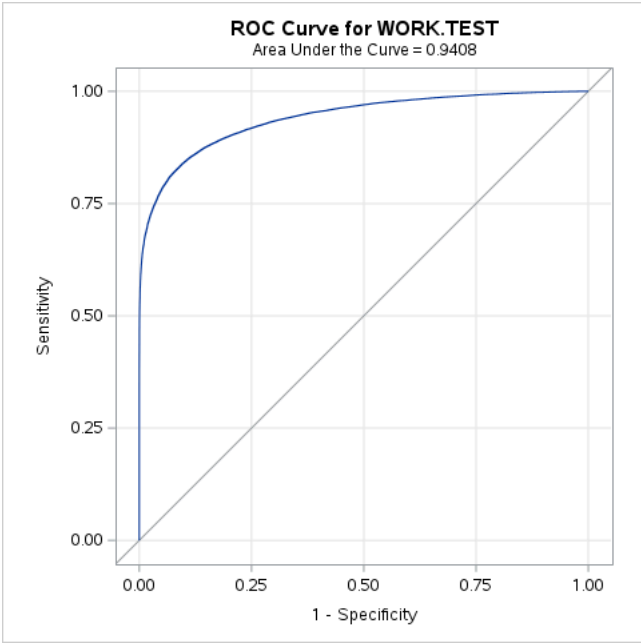
PaymentRisk =	0.33168 * Support Calls + 0.89143 * Payment Delay
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Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	2.2028	0.0239	8516.3278	<.0001
Age	1	0.0658	0.0131	25.0801	<.0001
Tenure	1	-0.1054	0.00644	268.2239	<.0001
Usage Frequency	1	-0.1174	0.00549	457.0607	<.0001
Support Calls	1	2.1556	0.0103	43666.3915	<.0001
Payment Delay	1	0.8625	0.00630	18715.3989	<.0001
Total Spend	1	-1.3456	0.00737	33306.0159	<.0001
Last Interaction	1	0.3605	0.00554	4226.8391	<.0001
SupportCallRate	1	-0.0469	0.0144	10.5496	0.0012
SpendingPerTenure	1	0.0470	0.00892	27.7839	<.0001
PaymentRisk	0	0	.	.	.
SubType_Enc	1	-0.0678	0.00675	101.1277	<.0001
Contract_Enc	1	-1.3226	0.00872	22985.3463	<.0001
AgeGroup_Enc	1	0.5862	0.0192	930.1242	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
Age	1.068	1.041	1.096
Tenure	0.900	0.889	0.911
Usage Frequency	0.889	0.880	0.899
Support Calls	8.633	8.460	8.810
Payment Delay	2.369	2.340	2.399
Total Spend	0.260	0.257	0.264
Last Interaction	1.434	1.419	1.450
SupportCallRate	0.954	0.928	0.982
SpendingPerTenure	1.048	1.030	1.067
SubType_Enc	0.934	0.922	0.947
Contract_Enc	0.266	0.262	0.271
AgeGroup_Enc	1.797	1.731	1.866

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	94.0	Somers' D	0.881
Percent Discordant	6.0	Gamma	0.881
Percent Tied	0.0	Tau-a	0.432
Pairs	30535244466	c	0.940





The CONTENTS Procedure

Data Set Name	WORK.PRED_TRAIN	Observations	352667
Member Type	DATA	Variables	22
Engine	V9	Indexes	0
Created	05/09/2026 00:52:45	Observation Length	184
Last Modified	05/09/2026 00:52:45	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label	Posterior Probabilities for DATA=WORK.TRAIN.		
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	497
First Data Page	1
Max Obs per Page	711
Obs in First Data Page	685
Number of Data Set Repairs	0
Filename	/saswork/SAS_workB5F70000EF99_odaws02-usw2-2.oda.sas.com/SAS_work6F130000EF99_odaws02-usw2-2.oda.sas.com/pred_train.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2281705506
Access Permission	rw-r--r--
Owner Name	u64508356
File Size	62MB
File Size (bytes)	65273856

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Informat	Label
2	Age	Num	8	BEST12.	BEST32.	
18	AgeGroup_Enc	Num	8			
9	Churn	Num	8	BEST12.	BEST32.	
14	Contract_Enc	Num	8			
1	CustomerID	Num	8			
19	F_Churn	Char	12			From: Churn
10	Gender_Female	Num	8			
11	Gender_Male	Num	8			
12	Gender_Unknown	Num	8			
20	I_Churn	Char	12			Into: Churn
8	Last Interaction	Num	8	BEST12.	BEST32.	
22	P_0	Num	8			Predicted Probability: Churn=0
21	P_1	Num	8			Predicted Probability: Churn=1
6	Payment Delay	Num	8	BEST12.	BEST32.	
17	PaymentRisk	Num	8			
16	SpendingPerTenure	Num	8			
13	SubType_Enc	Num	8			
5	Support Calls	Num	8	BEST12.	BEST32.	

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
15	SupportCallRate	Num	8			
3	Tenure	Num	8	BEST12.	BEST32.	
7	Total Spend	Num	8	BEST12.	BEST32.	
4	Usage Frequency	Num	8	BEST12.	BEST32.	

The CONTENTS Procedure

Data Set Name	WORK.PRED_TEST	Observations	88166
Member Type	DATA	Variables	22
Engine	V9	Indexes	0
Created	05/09/2026 00:52:45	Observation Length	184
Last Modified	05/09/2026 00:52:45	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label	Posterior Probabilities for DATA=WORK.TEST.		
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	125
First Data Page	1
Max Obs per Page	711
Obs in First Data Page	685
Number of Data Set Repairs	0
Filename	/saswork/SAS_workB5F70000EF99_odaws02-usw2-2.oda.sas.com/SAS_work6F130000EF99_odaws02-usw2-2.oda.sas.com/pred_test.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	2281705508
Access Permission	rw-r--r--
Owner Name	u64508356
File Size	16MB
File Size (bytes)	16515072

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
2	Age	Num	8	BEST12.	BEST32.	
18	AgeGroup_Enc	Num	8			
9	Churn	Num	8	BEST12.	BEST32.	
14	Contract_Enc	Num	8			
1	CustomerID	Num	8			
19	F_Churn	Char	12			From: Churn
10	Gender_Female	Num	8			
11	Gender_Male	Num	8			
12	Gender_Unknown	Num	8			
20	I_Churn	Char	12			Into: Churn
8	Last Interaction	Num	8	BEST12.	BEST32.	
22	P_0	Num	8			Predicted Probability: Churn=0
21	P_1	Num	8			Predicted Probability: Churn=1
6	Payment Delay	Num	8	BEST12.	BEST32.	
17	PaymentRisk	Num	8			
16	SpendingPerTenure	Num	8			
13	SubType_Enc	Num	8			
5	Support Calls	Num	8	BEST12.	BEST32.	
15	SupportCallRate	Num	8			
3	Tenure	Num	8	BEST12.	BEST32.	
7	Total Spend	Num	8	BEST12.	BEST32.	
4	Usage Frequency	Num	8	BEST12.	BEST32.	

Obs	Cutoff	Sensitivity	Specificity	Youden_Index
1	0.58776	0.83830	0.90442	0.74273
2	0.58676	0.83860	0.90411	0.74271
3	0.58778	0.83828	0.90442	0.74271
4	0.58687	0.83854	0.90416	0.74271
5	0.58771	0.83830	0.90440	0.74270

The FREQ Procedure

Frequency

Table of Churn by Pred_Churn			
Churn	Pred_Churn		
	0	1	Total
0	131665	21041	152706
1	26522	173439	199961
Total	158187	194480	352667

The FREQ Procedure

Frequency

Table of Churn by Pred_Churn			
Churn	Pred_Churn		
	0	1	Total
0	32931	5196	38127
1	6601	43437	50038
Total	39532	48633	88165
Frequency Missing = 1			

Obs	Dataset	TP	TN	FP	FN	Recall	Precision	F1	Accuracy
1	Train	173439	131665	21041	26522	86.74%	89.18%	87.94%	86.51%
2	Test	43437	32931	5196	6601	86.81%	89.32%	88.04%	86.62%

Diff_Accuracy	Diff_Recall	Diff_Precision	Diff_F1
-0.00106	-0.00072	-0.00135	-0.00102

Obs	Dataset	TP	TN	FP	FN	Recall	Precision	F1	Accuracy
1	Tree_Train	155181	152706	0	44780	77.61%	100.0%	87.39%	87.30%
2	Tree_Test	38829	38127	0	11209	77.60%	100.0%	87.39%	87.29%